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Summary

Software Engineering student shipping full-stack apps in **Python**, **TypeScript** and **Java**: **React**/Next.js frontends, **REST APIs** and **PostgreSQL** (hand-written **SQL**, migrations). Applies **AI/ML** in production (an **LLM tool-calling agent**, a **constraint-programming** generator, tooling for **AI coding agents**) and ships with **Docker** and **CI/CD**.

Education

York University, Bachelor of Engineering in Software Engineering *Expected Dec. 2027*
Coursework: Data Structures and Algorithms, Object-Oriented Programming (Java), Operating Systems, Database Systems

Projects

z9bra: Daily Logic Puzzle | *Next.js, TypeScript, PostgreSQL, Python, OR-Tools* Live | GitHub | Jun 2026

- Shipped a Wordle-style daily logic puzzle in **Next.js** and **TypeScript** that reached **200+ players** and a peak of **60 daily actives**, with date-seeded grids so every player gets the same puzzle.
- Wrote the **Python** generator (open source) that uses **CP-SAT** to prove each puzzle has **exactly one solution**.
- Synced streaks across devices through Google OAuth and a **PostgreSQL** (Neon) data layer.

OverTerm: Terminal for AI Coding Agents | *Rust, Tauri v2, TypeScript, GitHub Actions* GitHub | Oct 2026

- Built a native macOS terminal in **Rust + Tauri v2** that detects whether an **AI coding agent** is running, finished or waiting, from a **vt100 screen model** of each pseudoterminal.
- Released it with a **one-command install** and a tag-triggered **GitHub Actions** pipeline that bumps the Homebrew cask; merged **7 pull requests from 2 other contributors**.
- Root-caused two misdiagnosed bugs by reproducing them in the running app: a window resize that dropped keyboard focus, and a global `write()` that sent every terminal's replies to the active tab.

BundesPredict: Match Forecasting Service | *Python, FastAPI, PostgreSQL, Docker* Live | GitHub | Jul 2026

- Hand-wrote a **Dixon–Coles Poisson model** in SciPy and **backtested it over 1,827 matches** against bookmaker odds.
- Built a Dockerized **FastAPI + PostgreSQL** backend with **Alembic migrations**, idempotent **SQL upserts** for scheduled ingest, and a weekly **GitHub Actions** workflow that refits the model.
- Added an **LLM tool-calling agent** (Anthropic SDK) whose adjustments are **server-clamped** and logged; CI replays recorded transcripts, so tests need no API key.

Deximon: Pokémon TCG Platform | *Python, FastAPI, PostgreSQL, React, Docker* Live | GitHub | Summer 2026

- **Largest contributor on a 6-person team** (**66 of 140 commits**) to a live collector platform, merging through **GitHub pull requests** with **CI** on every PR.
- Built binder persistence end to end, from the **React** UI through a **FastAPI REST API** to a normalized **PostgreSQL** schema with unique and check constraints.
- **Reviewed 8 teammate pull requests** and shipped fixes, including an **N+1 query** on the marketplace feed (SQLAlchemy eager loading) and a WebSocket crash on dead sockets.

Lab Equipment Reservation System | *Java 17, JavaFX, JUnit 5, Maven* GitHub | Dec 2025

- Co-built a layered **Java** app (**Facade** over **DAO**) on a 5-person team, with the **State** pattern guarding booking transitions.
- Added **1,086 Randoop-generated** regression tests to the team's **JUnit 5** suite, plus JaCoCo coverage and PItest.

Skills

Languages: Python, TypeScript, JavaScript, Java, SQL, Rust, HTML/CSS
Databases: PostgreSQL, SQL, schema design, Alembic, SQLAlchemy
Web & APIs: React, Next.js, FastAPI, Django REST Framework, REST APIs, OAuth 2.0, JWT, WebSockets
AI & ML: LLM APIs (Anthropic SDK), tool-calling agents, PyTorch, scikit-learn, NumPy, SciPy, pandas, OR-Tools
Cloud & DevOps: Git, GitHub Actions CI/CD, Docker, AWS, Vercel, Railway, Linux | **Testing:** pytest, JUnit, Vitest
Spoken Languages: English (fluent), Turkish (fluent), Japanese (N4)